WH-1, 2 WATER HEATERS

1126 8th Street
Kirkland, WA 98033
P 425-576-0278
F 425-576-0748
Info@SpecSls.net

TO: OWP-LLC

ATTN: Richard Donelson

JOB NAME: 19 West Harrison Project

LOCATION: Seattle, Washington

Submittal Data

PVI INDUSTRIES, LLC

Item:	**Description: WH-1 & WH-2 High Efficient Water Heaters**
2 Each:	PVI #40 L 130A-CGL Conquest high efficiency condensing water heater 399,000 Btu, 460 gph, @ 100 degree rise 130 Gallon storage 80 3/4“ high x 33 3/4“ dia x 52“ deep, 4“ flue outlet duplex stainless steel tank. 99% thermal efficiency at part load. ASME Construction Modbus RTU connectable. Programmable electronic operating control. Digital temperature readouts, adjustable from 70F to 180F. Low NOx, less than 20 ppm, SCAQMD compliant. Equipped for direct combustion air connection. Vents with PVC, CPVC, or Polypropylene, up to 100 equivalent feet. 15-year tank warranty.

Enclosures: See Attached

WH-1, 2 WATER HEAT

CONQUEST®

Condensing Water Heater

Featuring...

AquaPLEX®
ENGINEERED DUPLEX ALLOY

15- year warranty.
Requires no tank linings
or heat exchanger linings.
Requires no anodes of any type.

UP TO 97% THERMAL EFFICIENCY AT FULL RATE FROM 40°F TO 140°F

TANK AND HEAT EXCHANGER MADE FROM AquaPLEX® ENGINEERED DUPLEX STAINLESS STEEL ALLOY

SEAMLESS VFD MODULATION REDUCES CYCLING AND IMPROVES EFFICIENCY UP TO 99% DURING LOW LOAD CONDITIONS

399 • 500 • 600
700 • 800 MBH INPUTS

130 GALLON TANK

EXTREMELY QUIET OPERATION

c ETL us Intertek

ASME HLW

ASHRAE 90.1 - 2010 compliant
SCAQMD compliant

PVI®
Engineered Water Heating Solutions®

WH-1, 2 WATER HEATERS

CONQUEST®

Condensing Water Heater

With a tank and heat exchanger constructed of AquaPLEX® duplex stainless steel alloy, the Conquest water heater combines an advanced fuel saving design with the extended product life long desired by the owners of condensing equipment.

Corrosion Resistance

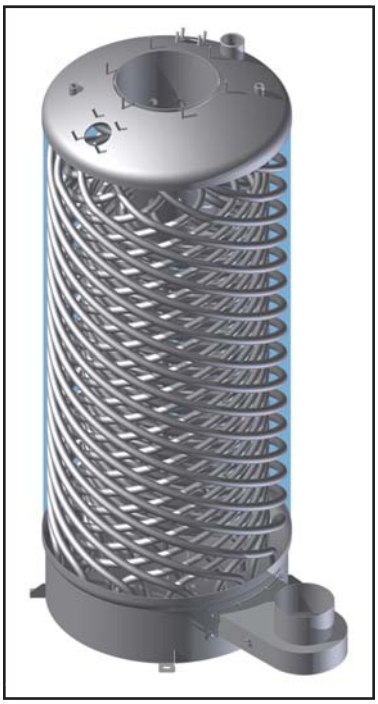
AquaPLEX is inherently corrosion resistant and entirely eliminates the need for a tank lining and anode rods, whether sacrificial or impressed current. Because there's no corrosion, there's simply nothing for an anode rod to do. And unlike 316L stainless steel, AquaPLEX is immune to chloride stress corrosion cracking, a known failure mode of 316L in potable water.

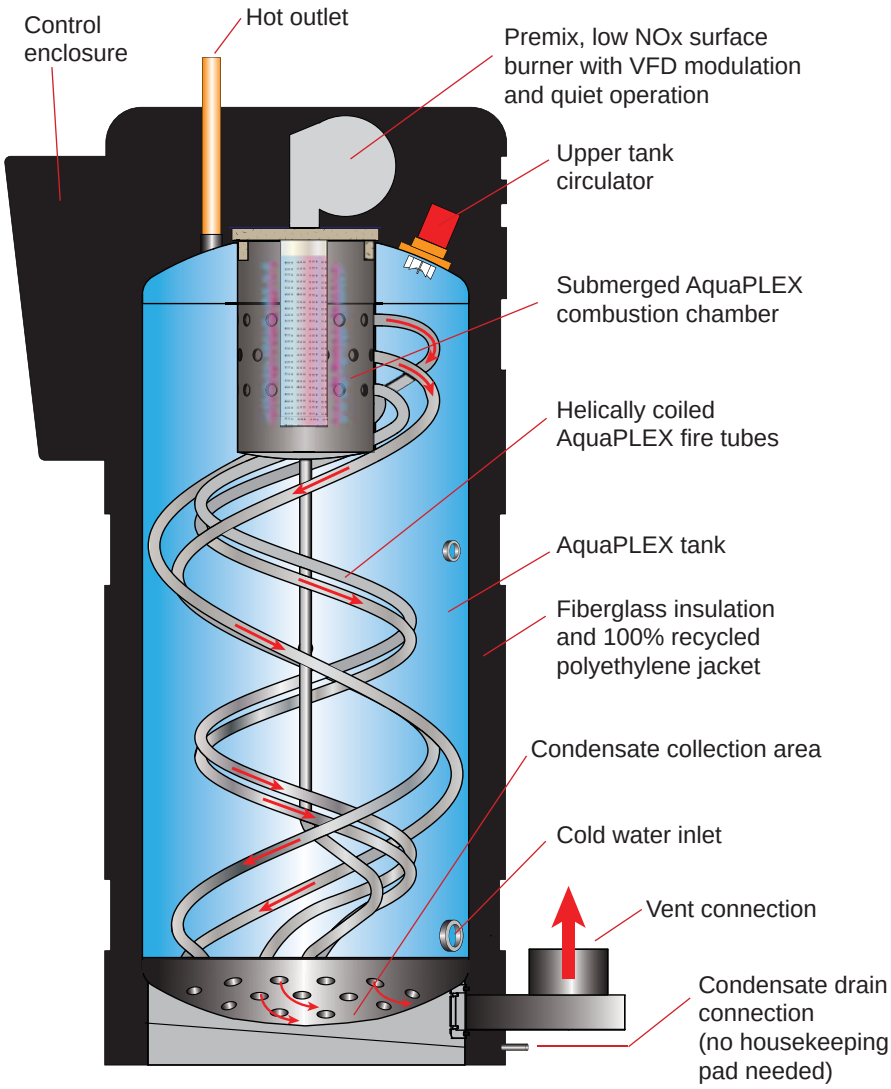
Condensing Efficiency

Efficiency is accomplished through a down-fired design beginning with a submerged combustion chamber and continuing through an array of helical fire tubes. Combustion gases are counter-flow to the direction of the potable water. This enables the coolest flue gases to contact the coldest water and raises low-fire efficiency to 99%.

AquaPLEX®
ENGINEERED DUPLEX ALLOY

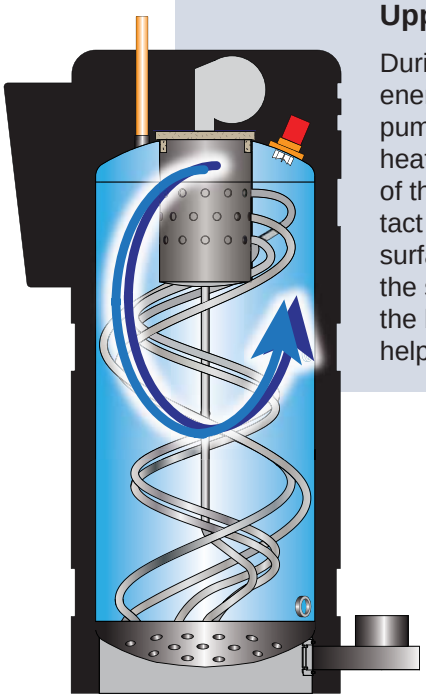
The pressure vessel, fire tubes and combustion chamber in the Conquest water heater are fabricated from AquaPLEX duplex alloy. AquaPLEX combines the grain structures of both 300 and 400 series stainless steels for unequalled corrosion protection.





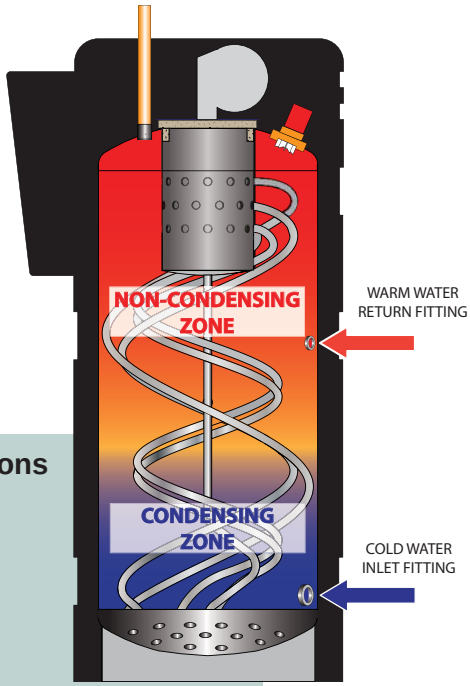
WH-1, 2 WATER HEATERS

ACHIEVES HIGH EFFICIENCY IN THE LAB AND MAINTAINS HIGH EFFICIENCY ON THE JOB



Upper Tank Circulator

During burner operation, Conquest energizes an integral circulating pump that forces water across the heating surfaces in the hottest part of the tank. More aggressive contact between the water and heating surfaces improves efficiency and the scouring action helps to reduce the buildup of scale. Circulation also helps to equalize tank temperature.

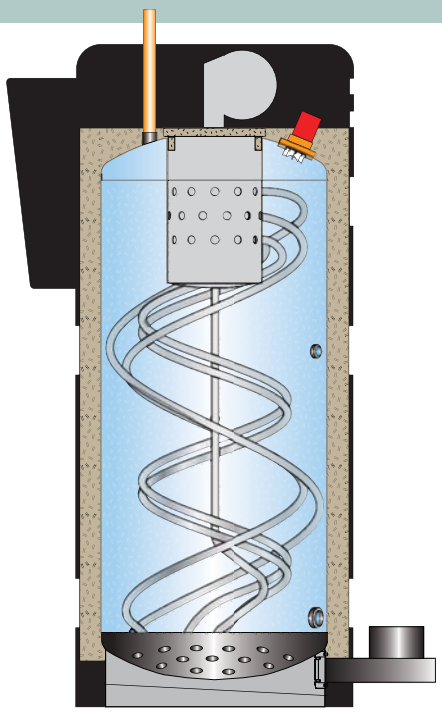


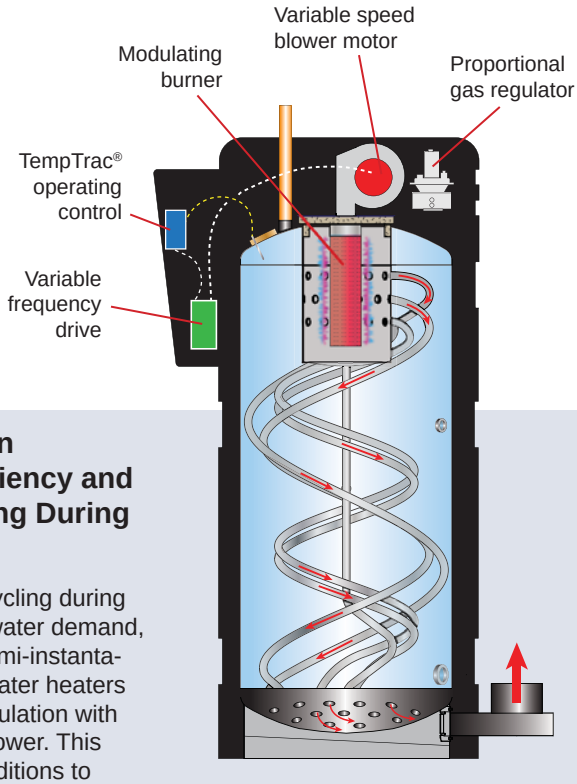
Both Warm and Cold Tank Connections

Constant circulation of hot water into the cold fitting on a condensing water heater lowers the efficiency. Conquest provides a dedicated fitting for connection to building return loops or side-arm tanks, maintaining two distinct temperature zones and allowing only the coldest water to enter the lower condensing zone of the water heater.

Low Standby Losses

Fiberglass insulation and a relatively small tank size reduce standby losses to levels 50% to 60% below ASHRAE 90.1 limits. Standby loss equates to an annual cost of about $40; less than the cost to run a small pump.





VFD Modulation Increases Efficiency and Reduces Cycling During Low Demand

To reduce burner cycling during periods of low hot water demand, the higher input (semi-instantaneous) Conquest water heaters employ burner modulation with a variable speed blower. This allows low flow conditions to be met with continuous, low BTU input and without short-cycling. The reduction in energy input also improves thermal efficiency up to 99%.

During periods when nominal demand is only a few gpm, the at-temperature storage capacity of the Conquest heater can meet the hot water requirement for 20 to 40 minutes before a burner cycle is needed.

WH-1, 2 WATER HEATERS

SELECTED STANDARD EQUIPMENT

- Up to 97% thermal efficiency at full fire from 40°F to 140°F
- Up to 99% thermal efficiency at low fire
- < 20 ppm NOx, SCAQMD listed
- Equipped for direct combustion air connection
- Vents through PVC or Polypropylene
- 15-year warranty for tank and exchanger

PRESSURE VESSEL & HEAT EXCHANGER

- AquaPLEX tank (unlined duplex alloy)
- AquaPLEX combustion chamber and fire tubes, helically coiled, single-pass, 100% submerged
- Temperature and pressure relief valve
- Fiberglass insulation
- 100% recycled, polyethylene jacket
- Drain valve

BURNER, OPERATING and SAFETY CONTROLS

- Pre-mix surface burner with VFD modulation and proportional gas/air control (≥ 500 MBH)
- UL and FM compliant gas train
- Programmable electronic operating control with digital temperature display embedded with Modbus RTU protocol
- Visual indication of modulation rate
- Electronic ignition control with pre-purge
- Manual-reset high limit control
- Electronic low-water cutoff
- Audible and visual alarm on any failure with contacts for remote notification
- Relay and proving contact for air louvers

CODES & STANDARDS

- Intertek /ETL listed to U.S. and Canadian standards
- Intertek /ELT listed for PVC or Polypropylene venting material, NSF-5, zero-clearance installation and low-lead compliance.
- ASHRAE 90.1 compliant

OPTIONAL EQUIPMENT

- CSD-1 controls (500 to 800 MBH)
- LP gas operation
- Condensate neutralization system
- Modbus cable for network communication
- Bacnet or Lonworks protocol gateways

Modbus Enabled Operating and Modulation Control

The TempTrac® electronic operating control allows the building's automation system to monitor and control the operation of the Conquest water heater through built-in Modbus RTU protocol. Network communicated points include operating set point (remotely adjustable), modulation rate and alarm status.

At the water heater, the control displays set point, sensed temperature, firing rate and alarm status. All parameters are fully programmable. Custom communication gateways are available for BacNet and Lonworks building automation systems.

Multiple Positive-Pressure Venting Options

Conquest is a category IV vented product listed for PVC and Polypropylene materials. Capable of sealed combustion with terminations in different pressure zones. Concentric venting is available on some sizes.

Conventional

Room air with sidewall vent

Sealed combustion with roof terminations

Sealed combustion with sidewall terminations

WH-1, 2 WATER HEATERS

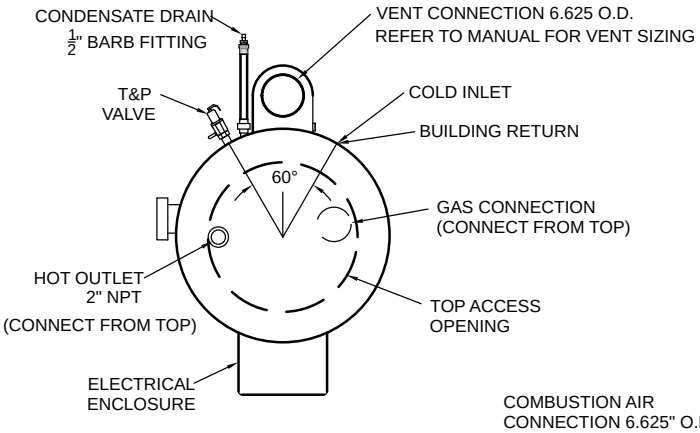
CONQUEST®

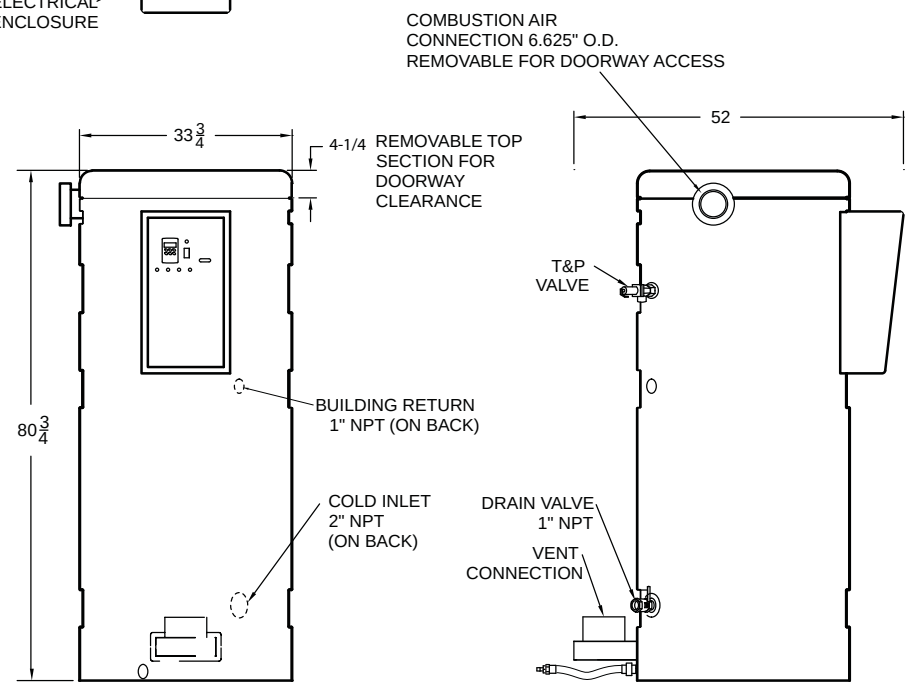
Condensing Water Heater

Model ▼	Input Btuh	Modulation	Recovery GPH (thermal efficiency)		Gas Connection NPT	Vent Diameter (ETL listed for longer vents with larger diameter)	Operating Weight (lbs.)
			70°F to 140°F ①	40°F to 140°F ②			
40 L 130A-GCL	399,000	No	657 (96%)	471 (97%)	3/4	4" @ 100 equivalent ft	1800
50 L 130A-GCML	500,000	Yes	823 (96%)	588 (97%)	1	4" @ 100 equivalent ft	
60 L 130A-GCML	600,000	Yes	988 (96%)	699 (97%)	1	6" @ 150 equivalent ft	
70 L 130A-GCML	700,000	Yes	1152 (96%)	815 (97%)	1	6" @ 150 equivalent ft	
80 L 130A-GCML	800,000	Yes	1317 (96%)	932 (97%)	1	6" @ 150 equivalent ft	

① Recoveries and thermal efficiency derived from DOE 10 CFR 431 testing requirements (ANSI Z21.10.3 @ 70°F to 140°F).
② Recoveries and thermal efficiency based upon 40°F entering water temperature.
Empty weight for all models is 680 lbs. Shipping weight is 880 lbs.

Standard Electric
120V, 1ø, 60 hz. (all models < 11 amps)
Gas Pressure (natural gas)
3.5" w.c. minimum flow pressure 14" w.c. maximum static pressure
Minimum clearances
Zero inches from sides and rear 15 inches from top Can be installed on combustible floor
Emissions
< 20 ppm NOx, SCAQMD compliant





FRONT VIEW

SIDE VIEW

PVI® *Engineered Water Heating Solutions®*

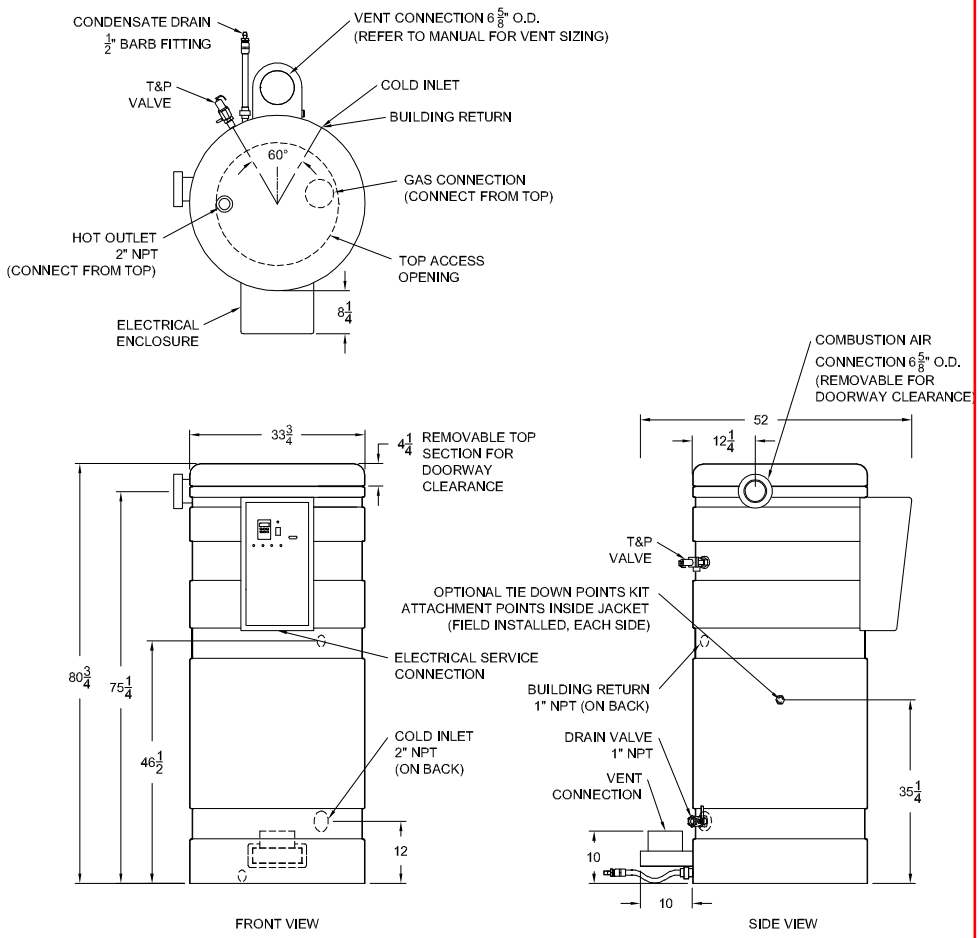
WH-1, 2 WATER HEATERS

Conquest®, CONDENSING WATER HEATER

AquaPLEX® Storage Tank and Heat Exchanger (unlined duplex alloy) • 130 Gallon Tank

MODEL ▼	INPUT BTU	Modulating	RECOVERY GPH (Thermal Efficiency)		GAS CONNECTION NPT	VENT DIAMETER (ETL listed for longer vents with larger dia.)	OPERATING Weight (lbs)
			70°F to 140°F ①	40°F to 140°F ②			
40 L 130A-GCL	399,000	No	657 (96%)	471 (97%)	3/4	4" @ 100 eq. ft.	1800
50 L 130A-GCML	500,000	Yes	823 (96%)	588 (97%)	1	4" @ 100 eq. ft.	
60 L 130A-GCML	600,000	Yes	988 (96%)	699 (97%)	1	6" @ 150 eq. ft.	
70 L 130A-GCML	700,000	Yes	1152 (96%)	815 (97%)	1	6" @ 150 eq. ft.	
80 L 130A-GCML	800,000	Yes	1317 (96%)	932 (97%)	1	6" @ 150 eq. ft.	

① Recoveries and thermal efficiency based upon DOE 10 CFR 431 testing per ANSI Z21.10.3 @ 70°F to 140°F.
② Recoveries and thermal efficiency based upon 40°F entering water temperature..
Dimensions are in inches unless otherwise indicated.
For standard and optional equipment, refer to form PV 8293.
Empty weight is 680 pounds and shipping weight is 880 pounds (all models)



Standard Electrical Service

120VAC, 1Ø, 60 Hz.
All models < 11 amps.

Venting

Use a Category IV PVC, CPVC or ETL, UL, ULC or CSA listed stainless steel or Centrotherm InnoFlue SW Polypropylene vent.
Minimum vent length is 5 eq. feet.
Maximum vent length varies by model number and vent diameter.
Do not size vent based only upon connection diameter at the appliance.
Refer to installation manual for sizing requirements.

Inlet Combustion Air Duct

Use PVC or galvanized pipe.
For model 40, up to 100 eq. ft. using 4" diameter pipe.
For model 50, up to 60 eq. ft. using 4" diameter pipe.
For models 60, 70 & 80, up to 150 eq. ft. using 6" pipe.
Longer lengths are ETL listed with larger diameters, refer to installation manual.

Gas Pressure - Natural

Minimum inlet flow pressure 3.5" W.C.
Maximum static pressure 10.5" W.C.
For LP gas, refer to installation manual.

Minimum Clearance from Combustibles

Zero clearance from sides and rear, 24" from front, 15" from top. Can be installed directly on a combustible floor.

Recommended Service Clearances

18" from all sides.
Check local and national codes for additional clearance requirements.

Emissions

All models < 20 ppm NOx.

PVI INDUSTRIES, LLC
Engineered Water Heating Solutions®
www.pvi.com
Fort Worth, TX ▪ 800-784-8326

PVI RESERVES THE RIGHT TO CHANGE THE DESIGN AND SPECIFICATION WITHOUT NOTICE.

WH-1, 2 WATER HEATERS

15-YEAR AquaPLEX® TANK AND HEATING SURFACES LIMITED WARRANTY
Covers CONQUEST®, Power VT® Plus, TRICON and MAXIM 3 Water Heaters

WHAT THIS LIMITED WARRANTY COVERS

When used for heating or storing potable water for domestic water heating purposes, this limited warranty covers defects in materials or workmanship of the tank and/ or heating surfaces that result in any of the following:

- Production of Rusty or Discolored Water
- Leaking
- Chloride Stress Corrosion Cracking
- Leaking due to Fireside Condensate Corrosion (condensing water heaters only)

The external economizer used on Power VT Plus water heaters is warranted for defects in materials and workmanship for one year and is not covered by the terms of this warranty.

REPAIR LIMITED WARRANTY

For 8 years from the start of this limited warranty, if the tank or heating surfaces has a warranty-covered defect, PVI will repair the defect and bring the tank or heating surfaces back to working condition. If, in the judgment of PVI, repair is not practical, this repair limited warranty will not be in force and the replacement limited warranty will be in force.

REPLACEMENT LIMITED WARRANTY

For 15 years from the start of this limited warranty, if, in the judgment of PVI, a warranty-covered defect of the tank or heating surfaces is not repairable, PVI will replace the water heater with the then current equivalent model. The first 8 years will be at no cost to the owner for the water heater. No labor or freight will be authorized or paid by PVI.

From the beginning of the 9th year to the end of the 15th year from the start of this limited warranty, PVI will offer to sell the owner a new water heater of the then current equivalent model.

The cost to the owner for the replacement water heater will be equal to the number of months in service from the date of start of this limited warranty, divided by the total number of months of this limited warranty, times the then current list price. At no point will the cost to the owner for the replacement water heater exceed 75% of the current list price. No labor or freight will be authorized or paid by PVI.

START OF LIMITED WARRANTY

FOR PRODUCTS WHERE FACTORY-AUTHORIZED START -UP IS REQUIRED: This limited warranty will begin at the time of start-up or six months from the date of shipment, whichever occurs first.

FOR PRODUCTS WHERE FACTORY-AUTHORIZED START -UP IS NOT REQUIRED: This limited warranty will begin at receipt of shipment.

GENERAL CONDITIONS

Under no circumstances will PVI pay for, or be responsible for, overtime pay (nights, weekends, or holidays) for the owner's convenience or desires. Labor cost covered by this warranty is limited to installations with normal access to the equipment. All additional cost resulting from uncommonly restrictive ingress or egress requiring intricate rigging and/or unusual building or machinery alterations will be the owner's responsibility.

Installation and operation must comply with all applicable federal, state, and local regulations, and must be in accordance with applicable instructions and limitations contained in the PVI installation and maintenance manual for the product. Normal maintenance and repair shall be performed by the owner. This limited warranty does not cover abuse or misuse of the product. This limited warranty is void if modifications are made to the product without the knowledge and consent of PVI. This limited warranty does not cover failure resulting from the accumulation of precipitant inside the vessel. This limited warranty does not cover tanks or heaters used for non-domestic (process) applications or for non-potable or high-purity water. Separate warranties are provided for these applications.

THIS LIMITED WARRANTY IS IN LIEU OF ALL OTHER WARRANTIES AND GUARANTEES, WHETHER EXPRESSED OR IMPLIED, INCLUDING STATUTORY WARRANTIES, SUCH AS WARRANTIES OF MERCHANTABILITY OR FITNESS FOR A PARTICULAR PURPOSE. The sole and exclusive remedy is repair or replacement of defective products pursuant to the terms of this limited warranty. **IN NO EVENT SHALL PVI BE LIABLE FOR CONSEQUENTIAL OR INCIDENTAL DAMAGES, OR CONTINGENT LIABILITIES OF ANY KIND, RESULTING FROM THE MANUFACTURE, SALE, INSTALLATION, OR USE OF THIS PRODUCT. SOME STATES DO NOT ALLOW LIMITATIONS ON INCIDENTAL OR CONSEQUENTIAL DAMAGES OR ON THE LENGTH OF AN IMPLIED WARRANTY. THE ABOVE LIMITATIONS MAY NOT APPLY TO YOU. YOU MAY HAVE OTHER RIGHTS, WHICH VARY FROM STATE TO STATE.** In no event shall Seller's liability on any claim for damages arising out of or connected with the sales contract or the manufacture, sale, delivery, or use of the goods exceed the purchase price of the goods.

All legal actions for breach of this limited warranty, or the terms and conditions of the contract for sale of the product, must be commenced on or before the expiration of one year from the date of such breach. Otherwise, such action shall be considered barred by a period of limitations established by agreement. This limited warranty covers the first installation only. PVI shall be entitled to inspect the product prior to repair or replacement.

Warranty claims should be presented through prompt telephone notification to PVI at toll-free 1-800-433-5654.

PVI Industries, LLC • P.O. Box 7124 • Ft. Worth, TX 76111